Small-time Controllability of Control Systems on 3D Lie Groups

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Abstract

We study affine control systems with one control on 3D unimodular Lie groups. Such control systems are first classified into a finite number of equivalent systems. After completing this classification, we determine the small-time controllability of each of the equivalent systems.

1 Introduction

In [1], Agrachev, Kazandjian, and Pozzoli considered the Schrödinger equation for the quantum harmonic oscillator, given by

$$\begin{cases} i\partial_t \psi(t,x) &= \Big(-\frac{1}{2}\Delta + u_0(t)\frac{|x|^2}{2} + \sum_{j=1}^d u_j(t)x_j \Big)\psi(t,x) \\ \psi(t=0) &= \psi_0 \in L^2(\mathbb{R}^d, \mathbb{C}) \end{cases} \tag{1}$$

where the control $u = (u_0, \dots, u_d) \in PWC([0,T], \mathbb{R}^{d+1})$. This equation is highly relevant in quantum dynamics, with applications in physics and chemistry. Based on this equation, the paper studied the controllability of left-invariant control affine systems of the form

$$\frac{d}{dt}q(t) = q(t)\Big(a + u_0(t)b + \sum_{i=1}^d u_i(t)X_i + r(t)Z\Big) \tag{2}$$

where $q \in SL_2 \ltimes H_d$. Notabaly, the paper proved that equation (2) is small-time controllable on $SL_2 \ltimes H_d$, and that equation (2) with $u_i, r = 0$ for $i \in \{1, \dots, d\}$ is small-time controllable on SL_2.

This result motivates the question: does there exist other 3D unimodular Lie groups for which the statement is true? To answer this question, we study affine control systems with one control on 3D unimodular Lie groups, taking the form

$$\dot{q}(t) = (A + u(t)B)q(t) \tag{3}$$

with $q \in G$, $A, B \in \mathfrak{g}$, and u the control. G is a 3D unimodular Lie group, which is known to be completely classified into the Lie groups $SU(2)$, $SL(2)$, H, $SE(2)$, and $SH(2)$. For each, $\mathfrak{g}$ is the corresponding Lie algebra. In section 2, we first classify such systems on each of these Lie groups into normal forms, or in other words, equivalent systems. Apparently, there are only finitely many such systems. After this classification is done, we study controllability. We will begin section 3 by formally defining small-time controllability, among other concepts from geometric control theory. Then, we will characterise the small-time controllability of each of the normal forms found. This is a thorough process that will give an answer to the question posed.

The results of this paper are summarised in the following table. There are 26 different cases in total, of which two are small-time controllable.

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Lie group	Normal Forms	STC
$SU(2)$	(σ_1, σ_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(E+F, H)$	No
	$(E-F, H)$	Yes
	$(E, E-F)$	No
	(H, E)	No
	(F, E)	Yes
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	No
$SE(2)$	(Y, X)	No
	(Z, X)	No
	(X, Z)	No
$SH(2)$	(X, Z)	No
	(Y, Z)	No
	$(X+Y, Z)$	No
	$(X-Y, Z)$	No
	$(X, X+Y)$	No
	$(X, X-Y)$	No
	$(Z, X+Y)$	No
	$(Z, X-Y)$	No
	(Z, Y)	No
	(X, Y)	No
	(Z, X)	No
	(Y, X)	No

Table 1: Summary of Normal Forms and Small-Time Controllability

1.1 Convention

Recall the following definitions of the 3D unimodular Lie groups that will be used throughout the paper.

- $SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}$
- $SL(2) = \{X \in M_2(\mathbb{R}) : \det X = 1\}$
- $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$
- $SE(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$
- $SH(2) = \left\{ \begin{pmatrix} \cosh\theta & \sinh\theta & x \\ \sinh\theta & \cosh\theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$

The following are the corresponding Lie algebras and the basis elements chosen for each.

- $\mathfrak{su}(2)$: $\sigma_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $\sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\sigma_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$
- $\mathfrak{sl}(2)$: $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
- $\mathfrak{h}$: $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{se}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{sh}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

2 Classifying Control Systems

Consider the control system (3) with $q \in G$ and $A, B \in \mathfrak{g}$, as formulated in the introduction. When classifying (3) on a Lie group, we want to simplify the pair (A, B) into an equivalent pair, say $(\tilde{A}, \tilde{B})$, such that the original system is small-time controllable if and only if the equivalent system is small-time controllable. This equivalency is denoted $(A, B) \sim (\tilde{A}, \tilde{B})$.

The classification process makes use of the natural symmetries of the problem. The first are automorphisms of Lie algebras $\Phi : \mathfrak{g} \to \mathfrak{g}$, leading to $(A, B) \sim (\Phi(A), \Phi(B))$. In particular, inner automorphisms, which act by conjugation by some $g \in G$, give $(A, B) \sim (gAg^{-1}, gBg^{-1})$ when applied simultaneously on A and B. In some cases, we will also utilise outer automorphisms. Second, the control can be shifted by $\delta \in \mathbb{R}$, so

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

The control can further be scaled by a constant $\mu \in \mathbb{R}$, giving

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

Finally, the time can be scaled by a factor of $\lambda \in \mathbb{R}^*$ so that $(A, B) \sim (\lambda A, \lambda B)$. This last equivalence is not immediately obvious. To see this, first consider the $\lambda > 0$ case. Consider the trajectory $q(t)$ that is the solution to (3), and another trajectory $x(t) = q(\lambda t)$ resulting from scaling time by a factor of λ. Then

$$\dot{x}(t) = \frac{d}{dt}q(\lambda t) = \lambda\dot{q}(\lambda t) = (\lambda A + u(\lambda t)\lambda B)q(\lambda t) = (\lambda A + v(t)\lambda B)x(t)$$

Let the attainable set of (A, B) from q_0 from time $\tau \geq 0$ be $\mathcal{A}(\leq \tau, q_0)$. Let the attainable set of $(\tilde{A}, \tilde{B})$ from q_0 from time $T \geq 0$ be $\tilde{\mathcal{A}}(\leq T, q_0)$, where $T = \lambda\tau$. If (A, B) is small-time controllable, then $\mathcal{A}(\leq \tau, q_0) = G$ for every $q_0 \in G$ and $\tau > 0$. By choosing appropriate τ, we see that this is true if and only if $\tilde{\mathcal{A}}(\leq T, q_0) = G$ for every $q_0 \in G$ and $T > 0$. So (A, B) is small-time controllable if and only if $(\lambda A, \lambda B)$ is small-time controllable. For $\lambda < 0$ we build on the last result, and just need to further show that $(A, B) \sim (-A, -B)$. Considering the reversed trajectory $x(t) = q(T - t)$ and following the same strategy as before extends the equivalence to all $\lambda \in \mathbb{R}^*$. The equivalence relations are summarised in the following definition.

Definition 2.1. For the control system (3), we have the following equivalence relations.

1. Automorphisms of Lie algebras: $(A, B) \sim (\Phi(A), \Phi(B))$

2. Control shifting: $(A, B) \sim ((A + \delta B), B)$ for $\delta \in \mathbb{R}$

3. Control scaling: $(A, B) \sim (A, \mu B)$ for $\mu \in \mathbb{R}$

4. Time scaling: $(A, B) \sim (\lambda A, \lambda B)$ for $\lambda \in \mathbb{R}^*$

Assumption. We will assume that B is nonzero, otherwise the control cannot act on the system. We will also assume that A and B are independent, otherwise the system can only be driven in a single direction.

It is useful to combine points 3 and 4 of Definition 2.1 to obtain $(\mu A, \lambda B) \sim (A, B)$ whenever $\mu, \lambda \neq 0$, which is always true given the above assumption.

Furthermore, to complete the proof of the classification, we will take advantage of invariants to show that the normal forms derived are not equivalent to each other.

Similar work regarding the classification of control systems has already been partially done by Remsing, for example in [3, 4, 5], notabaly on the $SU(2)$, $SL(2)$, H, and $SE(2)$ groups. In any case, we will provide, along with its proof, a more complete classification in this paper.

Proposition 2.2. Control systems of the form (3) on each 3D unimodular Lie group are equivalent to one of the model systems for the 3D unimodular Lie group, as listed in Table 1.

Proof. The demonstration for this proposition is given throughout the following sections. □

2.1 Classifying Control Systems on $SU(2)$

We consider the system (3) with $q \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0\sigma_1 + a_2^0\sigma_2 + a_3^0\sigma_3, \ B = b_1^0\sigma_1 + b_2^0\sigma_2 + b_3^0\sigma_3.$$

Let $K(\cdot, \cdot)$ denote the Killing form. $\mathfrak{su}(2)$ endowed with the Killing form is isomorphic to $\mathbb{R}^3$ with the standard Euclidean scalar product, where the isomorphism is given by $\sigma_i \mapsto e_i$, $i \in \{1, 2, 3\}$. Inner automorphisms act as rotations in $\mathbb{R}^3$. Alternatively, we can see this by noting

$$Aut(\mathfrak{su}(2)) = SO(3)$$

So, we construct a 3D space with σ_1, σ_2, σ_3 as the axes, analogous to $\mathbb{R}^3$. Applying a combination of conjugation group actions to both A and B, we rotate B onto the σ_3 axis, then rotate A onto the σ_1-σ_3 plane.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1\sigma_1 + a_3^1\sigma_3, b_3^1\sigma_3) \\
&\sim (a_1^1\sigma_1, b_3^1\sigma_3) \\
&\sim (a_1^1\sigma_1, a_1^1\sigma_3) \\
&\sim (\sigma_1, \sigma_3)
\end{aligned}$$

Hence, all control systems (3) on $SU(2)$ are equivalent to (σ_1, σ_3).

2.2 Classifying Control Systems on $SL(2)$

We consider the system (3) with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. Recall that for matrices $u_i, u_j \in \mathfrak{sl}(2)$, the matrix of the Killing form is the matrix K whose entries are $k_{ij} = K(u_i, u_j)$. Using the aforementioned basis elements, we compute the Killing form matrix to be

$$K = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

The eigenvalues of this matrix are -4, 4, and 8. There are 2 positive eigenvalues and 1 negative eigenvalue, implying a metric signature $(2,1)$. So the inner automorphisms of $SL(2)$ are transformations in Lorentzian space $\mathbb{R}^{2,1}$. The Killing form is analogous to the Minkowski scalar product.

In Lorentzian space, vectors can be classified as elliptic, hyperbolic, or parabolic based on the sign of their norm. Let $X \in \mathfrak{sl}(2)$. If $K(X,X) = 0$, then X is parabolic, lying on the light cone. If $K(X,X) < 0$, then X is elliptic, lying on the hyperboloid of two sheets inside the light cone. If $K(X,X) > 0$, then X is hyperbolic, lying on the hyperboloid of one sheet outside the light cone.

Lorentz transformations - including null rotations, rotations, and Lorentz boosts - act via inner automorphisms. There does not exist a Lorentz transformation between the hyperboloid of two sheets and the hyperboloid of one sheet. There also does not exist a Lorentz transformation between the upper and lower sheets of the hyperboloid of two sheets.

The strategy is to identify a representative for each class. We choose E, H, $E-F$ as the representatives for the parabolic, hyperbolic, and elliptic cases, respectively. B could be in any of these classes, giving rise to three cases to be analysed. Since this representative and B lie in the same region of space, we can apply Lorentz transformations via conjugation to simplify B to this representative. We may then proceed to simplify A.

Remark. If, in our convention, E is the representative for the upper half of the light cone, then $-E$ is the representative for the lower half of the light cone. The same goes for the hyperboloid of two sheets, by taking $E-F$ and $F-E$. However, it is not necessary to discuss the upper and lower halves separately. One can be transformed to the other by scaling the control.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 E + a_2^0 F + a_3^0 H, \ B = b_1^0 E + b_2^0 F + b_3^0 H.$$

Case 1: $K(B,B) > 0$, B is hyperbolic

By a Lorentz transformation, B can be simplified to a multiple of H.

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}(A,B) &= (a_1^1 E + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 E, b_3^1 H) \\ &\sim (a_1^1 E, a_1^1 H) \\ &\sim (E, H)\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A,B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_2^1 F, b_3^1 H) \\ &\sim (a_2^1 F, a_2^1 H) \\ &\sim (F, H)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then we need to find elements in the stabiliser group of H to continue simplifying while preserving H. Such elements of $Stab(H)$ must be of the form

$$g \in \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} : ad = 1 \right\}$$

which represents a Lorentz boost. Applying to $(a_1^1E + a_2^1F + a_3^1H, b_3^1H)$, we have

$$\begin{aligned}(A, B) &\sim (a_1^1E + a_2^1F + a_3^1H, b_3^1H)\\ &\sim (a_1^1a^2E + a_2^1d^2F + a_3^1H, b_3^1H)\\ &\sim (a_1^1a^2E + a_2^1d^2F, b_3^1H)\end{aligned}$$

If $sgn(a_1^1) = sgn(a_2^1)$, then, using $ad = 1$, it is possible to choose $a, d \in \mathbb{R}$ such that $a_1^1a^2 = a_2^1d^2 = c$, with $c \in \mathbb{R}$ being a constant with the appropriate sign. It follows that

$$\begin{aligned}(A, B) &\sim (c(E + F), b_3^1H)\\ &\sim (c(E + F), cH)\\ &\sim (E + F, H)\end{aligned}$$

If $a_1^1 > 0$, $a_2^1 < 0$, then we can choose $a, d \in \mathbb{R}$ such that $a_1^1a^2 = -a_2^1d^2 = c$, with $c > 0$. Then

$$\begin{aligned}(A, B) &\sim (c(E - F), b_3^1H)\\ &\sim (c(E - F), cH)\\ &\sim (E - F, H)\end{aligned}$$

In the case $a_1^1 < 0$ and $a_2^1 > 0$, following the same procedure gives $(-E + F, H)$, which is equivalent to $(E - F, H)$.

Case 2: $K(B, B) < 0$, B is elliptic

By a Lorentz transformation, B can be simplified to a multiple of $E - F$.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1})\\ &= (a_1^1E + a_2^1F + a_3^1H, b_1^1(E - F))\end{aligned}$$

To continue, we need elements of the stabiliser group $Stab(E - F)$. These elements are

$$g \in \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a^2 + b^2 = 1 \right\}$$

which is a rotation matrix, and so the Lorentz transformations in this case act as rotations in a plane. If we fix $E - F$ as one axis, then for a 3D space, we need to construct two more, and it is on the plane formed by these two axes that we will rotate. Consider

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \; H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

By computing the Killing forms, one can check that $E-F$, $E+F$, and H are indeed orthogonal, forming a set of coordinate axes. Finding these two remaining axes can be thought of as finding the orthogonal complement of B, $B^\perp$, spanned by $E + F$ and H. Now we rewrite A and B in terms of this new set of coordinates, reduce A onto $B^\perp$ by control shifting, then rotate on $B^\perp$. That is,

$$\begin{aligned}(A, B) &\sim (a_1^1E + a_2^1F + a_3^1H, b_1^1(E - F))\\ &\sim (a_1^2(E + F) + a_3^2H, b_1^1(E - F))\\ &\sim (a_1^3(E + F), b_1^1(E - F))\\ &\sim (a_1^3(E + F), a_1^3(E - F))\\ &\sim (2a_1^3E, a_1^3(E - F))\\ &\sim (2a_1^3E, 2a_1^3(E - F))\\ &\sim (E, E - F)\end{aligned}$$

Remark. We could have equivalently rotated to H, which would give $(H, E-F)$. It does not matter which one we rotate to. Indeed, we always have $K(A,A) > 0$, hence rotations are all occuring on the hyperboloid of one sheet.

Case 3: $K(B,B) = 0$, B is parabolic

By a Lorentz transformation, B can be reduced to a multiple of E.

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1})\\ &= (a_1^1E + a_2^1F + a_3^1H, b_1^1E)\end{aligned}$$

Again, we find elements of the stabiliser group $Stab(E)$. Such elements are of the form

$$g \in \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} : a^2 = 1 \right\}$$

representing the null rotation. Applying this to $(a_1^1E + a_2^1F + a_3^1H, b_1^1E)$,

$$(A,B) \sim ((a_1^1 - b^2a_2^1 - 2aba_3^1)E + a_2^1F + (aba_2^1 + a_3^1)H, b_1^1E)$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A,B) &\sim (a_3^1H, b_1^1E)\\ &\sim (a_3^1H, a_3^1E)\\ &\sim (H,E)\end{aligned}$$

If $a_2^1 \neq 0$, then choose $b = -a_3^1/aa_2^1$ to eliminate the H component in A, giving

$$\begin{aligned}(A,B) &\sim (a_2^1F, b_1^1E)\\ &\sim (a_2^1F, a_2^1E)\\ &\sim (F,E)\end{aligned}$$

2.3 Classifying Control Systems on H

We consider the system (3) with $q \in H$ and $A, B \in \mathfrak{h}$. We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0X + a_2^0Y + a_3^0Z, \; B = b_1^0X + b_2^0Y + b_3^0Z$$

The Heisenberg group is different from the previous two scenarios in that the Killing form is degenerate. Firstly, we may use inner automorphisms. These preserve the X and Y components, only shifting the Z component. Taking

$$g = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

and applying this transformation to A and B, we obtain

$$gAg^{-1} = a_1^0X + a_2^0Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, \; gBg^{-1} = b_1^0X + b_2^0Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

The second type of important transformations are symplectic maps, which, in fact, belong to the outer automorphisms. The symplectic map uses a matrix $T \in Sp(2,\mathbb{R})$ to act by left matrix multiplication on the (X,Y) vector in A and B, and preserves the Z component. Symbolically, we can write the mapping $((X,Y),Z) \mapsto (T(X,Y),Z)$. Note that in the $n=2$ case, $Sp(2,\mathbb{R}) = SL(2,\mathbb{R})$.

Consider also the centre of the Lie algebra $\mathfrak{h}$, which is $\mathfrak{z}(\mathfrak{h}) = \{cZ : c \in \mathbb{R}\}$. We try to simplify B first, and to do so, we analyze two cases: either $B \in \mathfrak{z}(\mathfrak{h})$, or $B \notin \mathfrak{z}(\mathfrak{h})$.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 X + b_2^0 Y = 0$, i.e. $b_1^0 = b_2^0 = 0$. Employ a symplectic matrix

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \ ad - bc = 1$$

where $c = -a_2^0$, $d = a_1^0$, and a, b are chosen to satisfy $ad - bc = 1$. This symplectic map transforms the X and Y components of A into X. Therefore,

$$\begin{aligned}
(A,B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_3^0 Z) \\
&\sim (X + a_3^0 Z, b_3^0 Z) \\
&\sim (X, b_3^0 Z) \\
&\sim (X, Z)
\end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $(b_1^0, b_2^0) \neq (0,0)$. b_3^0 may or may not be 0. In the inner automorphism on B,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0) Z$$

Since $(b_1^0, b_2^0) \neq (0,0)$, we can choose $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B. That is,

$$\begin{aligned}
(A,B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z, b_1^0 X + b_2^0 Y) \\
&\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z, b_1^1 X) \\
&\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z, b_1^1 X)
\end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}
(A,B) &= ((-ba_1^0 + aa_2^0 + a_3^0) Z, b_1^1 X) \\
&= ((-ba_1^0 + aa_2^0 + a_3^0) Z, (-ba_1^0 + aa_2^0 + a_3^0) X) \\
&= (Z, X)
\end{aligned}$$

If $a_2^1 \neq 0$, then we take another $h \in H$ that is also an element of the stabiliser of X. This is

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in Stab(X)$$

We now apply the inner automorphism.

$$\begin{aligned}
(A,B) &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z, b_1^1 X) \\
&\sim (h(a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0) Z) h^{-1}, h(b_1^1 X) h^{-1}) \\
&= (a_2^1 Y + (a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0)) Z, b_1^1 X)
\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $c' \in \mathbb{R}$ so that $a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A,B) &\sim (a_2^1 Y, b_1^1 X)\\ &\sim (a_2^1 Y, a_2^1 X)\\ &\sim (Y,X)\end{aligned}$$

Now, we need to show that the three normal forms found above are non-equivalent. To do this, we identify invariants, that is the algebraic characteristics that remain unchanged under the elementary transformations. Starting with $\tilde{B}$, an invariant is the property $\tilde{B} \in \mathfrak{z}(\mathfrak{h})$, since inner and outer automorphisms both preserve the centre, and scaling also preserves the centre. So (X,Z) is not equivalent to (Z,X) or (Y,X). It remains to distinguish between these latter two cases. We move to $\tilde{A}$. An invariant is the property that the coefficient of Z is zero, or in other words, $\tilde{A} \in span\{X,Y\}$. Indeed, inner and outer automorphisms, scaling by a constant, and adding $\tilde{B}$ (which has zero Z coordinate here) all preserve this property. So (Z,X) are (Y,X) not equivalent. This proves that all three normal forms found are non-equivalent.

2.4 Classifying Control Systems on $SE(2)$

We consider the system (3) with $q \in SE(2)$ and $A, B \in \mathfrak{se}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z,\ B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SE(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cos\theta X + \sin\theta Y,\ gYg^{-1} = -\sin\theta X + \cos\theta Y,\ gZg^{-1} = yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. So on A and B, we have

$$gAg^{-1} = (a_1^0\cos\theta - a_2^0\sin\theta + a_3^0 y)X + (a_1^0\sin\theta + a_2^0\cos\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0\cos\theta - b_2^0\sin\theta + b_3^0 y)X + (b_1^0\sin\theta + b_2^0\cos\theta - b_3^0 x)Y + b_3^0 Z$$

There are two cases to consider: when $B \in Span(X,Y)$, and when $B \notin Span(X,Y)$.

Case 1: $B \in Span(X,Y)$

In this case, $B = b_1^0 X + b_2^0 Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since $b_3^0 = 0$, we can disregard x and y, and choose θ so that gBg^{-1} rotates B in the X-Y plane onto X, then simplify.

$$\begin{aligned}(A,B) &\sim (gAg^{-1}, gBg^{-1})\\ &= ((a_1^0\cos\theta - a_2^0\sin\theta + a_3^0 y)X + (a_1^0\sin\theta + a_2^0\cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X)\\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X)\\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X)\end{aligned}$$

If $a_3^0 \neq 0$, then we require elements of $Stab(X)$, so that conjugation preserves $b_1^1 X$. It is clear that we need $\theta = 0$. Moreover, noting the result of gZg^{-1}, we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A. Thus,

$$\begin{aligned}(A,B) &\sim (a_3^0 Z, b_1^1 X)\\ &\sim (a_3^0 Z, a_3^0 X)\\ &\sim (Z,X)\end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$ (otherwise we obtain the same result as before),

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Case 2: $B \notin Span(X, Y)$

Here, $b_3^0 \neq 0$. When applying inner automorphisms, we first choose $x = 0$ and $y = 0$ so that we only rotate in the X-Y plane.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos\theta - a_2^0 \sin\theta + a_3^0 y)X + (a_1^0 \sin\theta + a_2^0 \cos\theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z)\end{aligned}$$

If $b_1^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Apply a second inner automorphism, choosing $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to rotate onto X.

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

If $b_1^1 \neq 0$, choose another $g \in SE(2)$ with $\theta = 0$ so that the inner automorphism causes no rotation, only translation. Choosing $x = 0$, $y = -b_1^1/b_3^0$,

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X + b_3^0 Z)g^{-1}) \\ &= ((a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, (b_1^1 + b_3^0 y)X - b_3^0 xY + b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Finally, use one last inner automorphism and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ is chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X.

$$\begin{aligned}(A, B) &\sim (a_1^3 X, b_3^0 Z) \\ &\sim (a_1^3 X, a_1^3 Z) \\ &\sim (X, Z)\end{aligned}$$

Lastly, we prove that the above normal forms are not equivalent to each other. Beginning with $\tilde{B}$, an invariant is the property that $\tilde{B} \in [\mathfrak{se}(2), \mathfrak{se}(2)] = span\{X, Y\}$. This separates (Y, X) and (Z, X) from (X, Z). Then, for $\tilde{A}$, the invariant property we identify is the same as in the Heisenberg group case. Indeed, the property whether the Z direction vanishes is preserved under automorphisms, scaling, and adding $\tilde{B}$ (which does not contain Z in either of the two normal forms in question). From this invariant, it is clear that (Y, X) and (Z, X) are non-equivalent. Hence, all three normal forms found are non-equivalent.

2.5 Classifying Control Systems on $SH(2)$

Consider the system (3) with $q \in SH(2)$ and $A, B \in \mathfrak{sh}(2)$. The $SH(2)$ group is similar to $SE(2)$, except that $SH(2)$ is translations and hyperbolic rotations preserving the Lorentzian.

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \ B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cosh\theta X + \sinh\theta Y, \ gYg^{-1} = \sinh\theta X + \cosh\theta Y, \ gZg^{-1} = -yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. Indeed, these transformations are combinations of hyperbolic rotations determined by θ, and translations determined by x and y. For A and B, we have

$$gAg^{-1} = (a_1^0 \cosh\theta + a_2^0 \sinh\theta - a_3^0 y)X + (a_1^0 \sinh\theta + a_2^0 \cosh\theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y)X + (b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x)Y + b_3^0 Z$$

The analysis requires two cases to be considered: $B \in Span(X, Y)$ and $B \notin Span(X, Y)$.

Case 1: $B \in span\{X, Y\}$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$. We would like to begin by applying hyperbolic rotations to simplify B. Since $b_3^0 = 0$, to rotate B onto X or Y, we need to solve the equation $\tanh\theta = -b_2^0/b_1^0$ or $\tanh\theta = -b_1^0/b_2^0$. tanh is a mapping $\mathbb{R} \to (-1, 1)$, so at most one of these two equations can be solved, depending on the relative magnitudes $|b_1^0|$ and $|b_2^0|$. Three different cases are to be analysed.

If $|b_1^0| = |b_2^0|$, then none of the equations have a solution, and it is not possible to rotate B onto only X or only Y. We proceed as follows. If $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y)\\ &\sim (a_1^1 X, b_1^0 X + b_2^0 Y)\\ &= (a_1^1 X, b_1^0 (X \pm Y))\\ &\sim (a_1^1 X, a_1^1 (X \pm Y))\\ &\sim (X, X \pm Y)\end{aligned}$$

Equivalently, we could have extracted b_2^0 instead of b_1^0, giving the same result. A could have also been reduced to Y in the second line, which can be rotated to X. If $a_3^0 \neq 0$, apply a translation using a $g \in SH(2)$ with $\theta = 0$, $x, y \in \mathbb{R}$.

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y)\\ &\sim ((a_1^0 - a_3^0 y)X + (a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 (X \pm Y))\\ &= (a_3^0 Z, b_1^0 (X \pm Y))\\ &\sim (a_3^0 Z, a_3^0 (X \pm Y))\\ &\sim (Z, X \pm Y)\end{aligned}$$

We were able to translate A to only Z precisely because $a_3^0 \neq 0$, which allowed choices of y and x such that the coefficients of X and Y vanish. Again, we could have equivalently extracted b_2^0 instead of b_1^0.

If $|b_1^0| < |b_2^0|$, then $b_1^0 \cosh\theta + b_2^0 \sinh\theta = 0$ has a solution, and B can be rotated onto Y. If $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y, b_2^1 Y) \\ &\sim (a_1^1 X, b_2^1 Y) \\ &\sim (a_1^1 X, a_1^1 Y) \\ &\sim (X, Y)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_2^1 Y) \\ &\sim (a_1^1 X + a_3^0 Z, b_2^1 Y) \\ &\sim (a_3^0 Z, b_2^1 Y) \\ &\sim (a_3^0 Z, a_3^0 Y) \\ &\sim (Z, Y)\end{aligned}$$

where the fourth step is a translation by choosing $\theta = 0$, $x = 0$, $y = a_1^1/a_3^0$.

If $|b_1^0| > |b_2^0|$, then $b_1^0 \sinh\theta + b_2^0 \cosh\theta = 0$ has a solution, and B can be rotated onto X. If $a_3^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X)\end{aligned}$$

where the fourth step is a translation by choosing $\theta = 0$, $x = a_2^1/a_3^0$, $y = 0$.

Case 2: $B \notin span\{X, Y\}$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh\theta + b_2^0 \sinh\theta - b_3^0 y = 0$ and $b_1^0 \sinh\theta + b_2^0 \cosh\theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

We may assume $(a_1^1, a_2^1) \neq (0,0)$. Now, hyperbolic rotations will be applied with the intent of reducing A, instead of B. Regardless, the arguments from Case 1 still apply, and we aim to solve $\tanh\theta = -a_1^0/a_2^0$ or $\tanh\theta = -a_2^0/a_1^0$. We analyse three different cases based on the relative magnitudes $|a_1^1|$ and $|a_2^1|$.

If $|a_1^1| = |a_2^1|$, then neither of the two equations has a solution. So we proceed as

$$\begin{aligned}(A,B) &\sim (a_1^1(X \pm Y), b_3^0 Z)\\ &\sim (a_1^1(X \pm Y), a_1^1 Z)\\ &\sim (X \pm Y, Z)\end{aligned}$$

If $|a_1^1| < |a_2^1|$, then $a_1^1\cosh\theta + a_2^1\sinh\theta = 0$ has a solution and we can rotate onto Y. Note that we must choose $x = y = 0$ during the transformation so that $b_3^0 Z$ is preserved.

$$\begin{aligned}(A,B) &\sim (a_2^2 Y, b_3^0 Z)\\ &\sim (a_2^2 Y, a_2^2 Z)\\ &\sim (Y, Z)\end{aligned}$$

If $|a_1^1| > |a_2^1|$, then $a_1^1\sinh\theta + a_2^1\cosh\theta = 0$ has a solution and we can rotate onto X. Again choose $x = y = 0$ so that the $b_3^0 Z$ is preserved.

$$\begin{aligned}(A,B) &\sim (a_1^2 X, b_3^0 Z)\\ &\sim (a_1^2 X, a_1^2 Z)\\ &\sim (X, Z)\end{aligned}$$

3 Small-Time Controllability

Consider control systems of the form

$$\dot{q}(t) = (\tilde{A} + u\tilde{B})q(t) \tag{4}$$

where $q \in G$, and $\tilde{A}, \tilde{B} \in \mathfrak{g}$ are one of the normal forms listed in Table 1. For each of the normal forms found, we wish to determine its small-time controllability. Let us first recall some definitions, as given in [7].

Definition 3.1. Reachable (attainable) sets starting from q_0:

- the reachable set from q_0 at time $\tau \geq 0$ is

$$\mathcal{A}(\tau, q_0) = \{q_1 \in \mathbb{R}^n : \exists u(\cdot) : [0,\tau] \to U, q(\tau, q_0, u) = q_1\};$$

- the reachable set from q_0 within time $\tau \geq 0$ is

$$\mathcal{A}(\leq \tau, q_0) = \cup_{t\in[0,\tau]}\mathcal{A}(t, q_0);$$

- the reachable set from q_0 is

$$\mathcal{A}(q_0) = \cup_{t\in[0,\infty)}\mathcal{A}(t, q_0)$$

.

Definition 3.2. The system (4) is said to be

- controllable if for every $q_0 \in G$, $\mathcal{A}(q_0) = G$;
- small-time controllable if at the identity I, for every $\tau > 0$, we have $\mathcal{A}(\leq \tau, I) = G$;

- small-time locally controllable at q_0 if q_0 belongs to the interior of $\mathcal{A}(\leq \tau, q_0)$ for every $\tau > 0$.

We now state a classic theorem in control theory, as formulated in [8]. This theorem will be central in characterising small-time controllability.

Theorem 3.3 (Rashevsky-Chow)**.** Let M be a smooth manifold and $X^1, \ldots, X^m$ be m smooth vector fields on M. Assume that

$$Lie\{X^1, \ldots, X^m\}(x) = T_x M, \quad \forall x \in M$$

Then the control system

$$\dot{x} = \sum_{i=1}^{m} u_i X^i(x)$$

is locally controllable in arbitrary small time at every point of M.

Remark. If M is connected, then local controllability implies global controllability. A proof is given in [8]. This is useful since all the matrix Lie groups considered in this paper are connected.

We will study small-time controllability at the identity, that is, taking $q(0) = Id$. This is always possible by considering the translation mapping $q \mapsto qh$, where $h \in G$. This mapping translates the original trajectory to another trajectory with the same control.

We hope to apply the Rashevsky-Chow theorem to conclude small-time controllability. However, the problem is that the theorem applies only to driftless systems, while the system (4) clearly has a drift term $\tilde{A}$. Broadly speaking, the strategy to overcome this problem is to employ fast-oscillating controls and study the remaining drift. This will primarily be based on the works [2, 6].

First, we must check the bracket-generating condition. In the 3D case, it means whether

$$span\{\tilde{A}, \tilde{B}, [\tilde{A}, \tilde{B}]\} = \mathfrak{g}$$

Proposition 3.4. If the pair $(\tilde{A}, \tilde{B})$ is not bracket generating, then the system (4) is not small-time controllable.

Proof. Suppose $(\tilde{A}, \tilde{B})$ is not bracket generating, then the third direction needed to span the 3-dimensional Lie algebra cannot be generated, hence the pair is not small-time controllable. $\square$

On the other hand, if the bracket-generating condition is satisfied, then we may proceed to the next step. We consider the convexified adjoint orbit

$$conv(Ad_L \tilde{A})/\mathbb{R}\tilde{B}$$

where $L = \overline{\{e^{s\tilde{B}} : s \in \mathbb{R}\}}$, and study its geometry. We look for new drift directions in this orbit, characterised by an infinite line through the origin. If this new fast direction and the original fast direction generates the third direction needed to span the Lie algebra, then we are done. Otherwise, we repeat the above process, taking advantage of what is now effectively two fast directions, and so on. Evidently, the purpose of finding the convexified adjoint orbit is to search for new directions to use. Ideally, we hope to eventually obtain a system of the form

$$\dot{q}(t) = (u\tilde{A} + v\tilde{B} + w\tilde{C})q(t)$$

where $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ together span the Lie algebra $\mathfrak{g}$, and $(u(t), v(t), w(t)) \in \mathbb{R}^3$. This is an extended system that, as proven in [2, 6], is compatible with the original system (4), i.e. its attainable set is contained in the closure of the attainable set of (4). Notice there is no longer a drift term, but rather three fast directions. This puts us in a position to apply the Rashevsky-Chow theorem to conclude small-time controllability.

The following observation will be helpful for reducing repetitiveness when it comes to some frequently recurring geometries of the convexified set.

Proposition 3.5. If $ad^2_{\tilde{B}}(\tilde{A}) = 0$, then $conv(Ad_L \tilde{A})$ where $L = \{\overline{e^{s\tilde{B}}} : s \in \mathbb{R}\}$ is a straight line not passing through the origin, given by $\{\tilde{A} + s[\tilde{B}, \tilde{A}] : s \in \mathbb{R}\}$.

Proof. Let $ad^2_{\tilde{B}}(\tilde{A}) = 0$, which implies $[\tilde{B}, [\tilde{B}, \tilde{A}]] = 0$. Thus

$$\begin{aligned} Ad_{\overline{e^{s\tilde{B}}}} \tilde{A} &= e^{ad_{s\tilde{B}}}(\tilde{A}) \\ &= \tilde{A} + [s\tilde{B}, \tilde{A}] + \frac{1}{2}[s\tilde{B}, [s\tilde{B}, \tilde{A}]] + \cdots \\ &= \tilde{A} + s[\tilde{B}, \tilde{A}] \end{aligned}$$

Since the pair $(\tilde{A}, \tilde{B})$ is bracket generating, we can conclude

$$conv(Ad_L \tilde{A}) = \{\tilde{A} + s[\tilde{B}, \tilde{A}] : s \in \mathbb{R}\}$$

□

The above description is only for when we are able to successfully conclude small-time controllability. In practice, there are various points at which the method can terminate. If the set is bounded, we can immediately conclude that the system (4) is not small-time controllable.

Proposition 3.6. If the convexified adjoint orbit is bounded, then the system (4) is not small-time controllable.

Proof. For a system that is small-time controllable, it must be able to travel in any direction arbitrarily fast. This is clearly not possible when the convexified adjoint orbit is bounded. □

If the set is semi-infinite, then the situation becomes more delicate, and needs to be treated on a case-by-case basis. For instance, there are cases where it is preferable to follow another route to prove non-small-time controllability. The following method, inspired by barrier functions from convex optimisation, will be useful on numerous occasions.

Proposition 3.7. Assume there exists a barrier function F such that

1. $F(q(t)) > 0$ for all trajectories $q(t)$, $t > 0$, of the system (4),
2. $F(I) = 0$,
3. there exists a point X close to the identity I where $F(X) < 0$.

Then the system (4) is not controllable, and in particular, not small-time controllable.

Proof. Let F be a function that satisfies the three conditions above. Then there exists a point in a neighbourhood of the identity that cannot be reached by any trajectory of the system in positive time. So, the attainable set from the identity $\mathcal{A}(\leq \tau, I) \neq G$ for all $\tau > 0$. Hence the system is not small-time controllable. □

Proposition 3.8. The small-time controllability of control systems of the form (3) on each 3D unimodular Lie group is stated in Table 1.

Proof. The demonstration for this result is given throughout the following sections. □

3.1 Controllability on $SU(2)$

First, $[\sigma_1, \sigma_3] = -\frac{1}{2}\sigma_2$, so

$$span\{\sigma_1, \sigma_3, [\sigma_1, \sigma_3]\} = span\{\sigma_1, \sigma_3, -\frac{1}{2}\sigma_2\} = \mathfrak{su}(2)$$

The pair is bracket generating, so we may proceed to the next step. Define the set

$$L = \{\overline{e^{s\sigma_3}} : s \in \mathbb{R}\}$$

Note that in this case, the closure $\overline{e^{s\sigma_3}} = e^{s\sigma_3}$. It is easy to compute

$$Ad_L\sigma_1 = e^{s\sigma_3}\sigma_1 e^{-s\sigma_3} = \{\cos(s)\sigma_1 + \sin(s)\sigma_2 : s \in \mathbb{R}\}$$

We then convexify this set, and quotient out the fast direction σ_3.

$$conv(Ad_L\sigma_1)/\mathbb{R}\sigma_3 = \{(\sigma_1, \sigma_2) : \sigma_1^2 + \sigma_2^2 \leq 1\}$$

This set is a disc in the u_1-u_2 plane bounded by the unit circle. Since the convexified set is bounded, by Proposition 3.6, the system (σ_1, σ_3) is not small-time controllable.

3.2 Controllability on $SL(2)$

3.2.1 (E, H)

$[E, H] = -2E \in span\{E, H\}$, so by Proposition 3.4, the system is not small-time controllable.

3.2.2 (F, H)

$[F, H] = 2F \in span\{F, H\}$, so by Proposition 3.4, the system is not small-time controllable.

3.2.3 $(E + F, H)$

As $[E + F, H] = -2(E - F)$, it is easy to check that this system is bracket generating. Furthermore, the convexified adjoint orbit quotient the fast direction H is the region above one branch of a hyperbola in the E-F plane, expressed as

$$conv(Ad_{e^{sH}}(E + F))/\mathbb{R}H = conv\{e^{2s}E + e^{-2s}F : v \in \mathbb{R}\}$$

Such a set poses various obstacles, and instead of continuing along this route, it is considerably easier to employ the barrier function method instead. Consider the function

$$F = ac + bd$$

Now we transform the matrix representation into a state-space representation. From

$$\dot{q} = (E + F + uH)q$$

we see

$$\begin{cases} \dot{a} &= au + c \\ \dot{b} &= bu + d \\ \dot{c} &= -cu + a \\ \dot{d} &= -du + b \end{cases}$$

Using this, compute the derivative

$$\dot{F}(q(t)) = \dot{a}c + a\dot{c} + \dot{b}d + b\dot{d} = a^2 + b^2 + c^2 + d^2 \geq 0$$

In fact, given the restriction $ad - bc = 1$, it must be that

$$\dot{F}(q(t)) > 0$$

for all trajectories $g(t)$ of the system $(E + F, H)$. We can also check that at the identity,

$$F(I) = 1 \cdot 0 + 0 \cdot 1 = 0$$

Lastly, consider

$$\exp(-\varepsilon F) = \begin{pmatrix} 1 & 0 \\ -\varepsilon & 1 \end{pmatrix},$$

arbitrarily close to the identity. We have

$$F(\exp(-\varepsilon F)) = -\varepsilon < 0, \ \forall \varepsilon > 0$$

By Lemma 3.7, $(E + F, H)$ is not small-time controllable.

3.2.4 $(E - F, H)$

Since $[E - F, H] = -2(E + F)$, the pair $(E - F, H)$ is bracket generating. We proceed to find the convexified adjoint orbit.

$$conv(Ad_{e^{sH}}(E - F))/\mathbb{R}H = conv\{e^{2s}E - e^{-2s}F : s \in \mathbb{R}\}$$

This is the region above one branch of a hyperbola in the E-F plane. We attempt to go to the infinity in the $E + F$ direction within the convex hull. This is only the positive half of an infinite line through the origin. Consider the differential equation

$$\dot{q}(t) = (E - F + uH + v(E + F))q(t), \ u \in \mathbb{R}, v > 0$$

Choosing $u = 0$ and any $v \in \mathbb{R}$ such that $0 < v < 1$, the solution is

$$q(t) = [\cos(\sqrt{1 - v^2}t)I + \frac{\sin(\sqrt{1 - v^2}t)}{\sqrt{1 - v^2}}((v + 1)E + (v - 1)F)]q(0)$$

which is periodic. This suggests that the half line can be viewed as an infinite line in both directions along $E + F$, giving a new fast direction to use. Now we check the commutator, $[H, E + F] = 2(E - F) \notin span\{H, E + F\}$. H and $E + F$ successfully generated the third direction needed to span the Lie algebra. Finally, we have obtained the extended system

$$\dot{q}(t) = (u(t)(E - F) + v(t)H + w(t)(E + F))q(t)$$

which is compatible with the original system $(E - F, H)$. By the Rashevsky-Chow theorem, $(E - F, H)$ is small-time controllable.

3.2.5 $(E, E - F)$

$[E, E - F] = -H$, so

$$span\{E, E - F, [E, E - F]\} = \mathfrak{sl}(2),$$

and so the pair is bracket generating. We can proceed to find the convexified adjoint orbit, then quotient out the fast direction $E-F$. First, we compute the adjoint and write it in a favourable form.

$$\begin{aligned}
Ad_{\overline{e^{s(E-F)}}}E &= \{\cos^2(s)E - \sin^2(s)F + \sin(S)\cos(s)H : s \in \mathbb{R}\} \\
&= \{(\cos^2(s) - \sin^2(s))E + \sin^2(s)E - \sin^2(s)F + \sin(s)\cos(s)H : s \in \mathbb{R}\} \\
&= \Big\{ \cos(2s)E + \sin^2(s)(E-F) + \frac{1}{2}\sin(2s)H : s \in \mathbb{R}\Big\}
\end{aligned}$$

From this, the convexified adjoint orbit can be expressed explicitly.

$$conv(Ad_{\overline{e^{s(E-F)}}}E)/\mathbb{R}(E-F) = \{(E,H) : \cos(2E) + \frac{1}{2}\sin(2H) \leq 1\}$$

which describes the region bounded by an ellipse in the E-H plane. Hence, by Proposition 3.6, the system is not small-time controllable.

3.2.6 (H, E)

$[H, E] = 2E$, so by Proposition 3.4, the system is not small-time controllable.

3.2.7 (F, E)

$[F, E] = -H$, so the pair is bracket generating. We proceed to find the convexified adjoint orbit and quotient out the fast direction E. This is a straight line not passing through the origin given by

$$conv(Ad_{e^{vE}}F)/\mathbb{R}E = \{F + vH : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$conv(Ad_{e^{vE}}F) + uE = \{F + uE + vH : u, v \in \mathbb{R}\}$$

Going to the infinity along this convex hull, we can approximate an infinite line along H in both directions. We check the commutator $[H, E] = 2E \in span\{H, E\}$. E and H were unable to generate a new direction, so we need to continue to find quotient spaces, which gives us new directions to use. Now consider a modified system

$$\dot{q}(t) = (F + uE + vH)q(t)$$

This new system has two fast directions, E and H, which we quotient from the convexified adjoint orbit. This is done by computing

$$conv(Ad_{e^{uE+vH}}F)/span\{E, H\} = \{e^{-2v}F : v \in \mathbb{R}\}$$

This one-dimensional set is half of an infinite line in the positive F direction. Let us now check if the solution to the system

$$\dot{q}(t) = (wF + uE + vH)q(t),\ w > 0$$

is periodic. If it is, then the periodicity allows this set to be viewed as an infinite line passing through the origin. Suppose we take $v = 0$, $u = -1$, $w = 1$ so that

$$wF + uE + vH = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The solution to this matrix differential equation is

$$\begin{aligned} q(t) &= \exp\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} t\right) q(0) \\ &= [\cos(t) I + \sin(t)(F - E)] q(0) \end{aligned}$$

which is indeed periodic. In fact, the solution will always be periodic given $v = 0$, $u = -w$ with $w > 0$. This suggests that the half line can be seen as an infinite line in both directions along F. Applying the Rashevsky-Chow theorem, we conclude that the system (F, E) is small-time controllable.

3.3 Controllability on H

3.3.1 (X, Z)

$[X, Z] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.3.2 (Z, X)

$[Z, X] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.3.3 (Y, X)

Since $[Y, X] = -Z$, the pair is clearly bracket generating. We proceed to compute the convexified adjoint orbit, and quotient out the fast direction X. This is a straight line not passing through the origin, given by

$$conv(Ad_{e^{vX}} Y)/\mathbb{R}X = \{Y + vZ : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$conv(Ad_{e^{vX}} Y) + uX = \{Y + uX + vZ : u, v \in \mathbb{R}\}$$

Now, the X and Z directions can be made arbitrarily fast by taking large controls u and v. Checking the bracket

$$[Z, X] = 0 \in span\{Z, X\},$$

we see that X and Z were unable to generate a new direction. Repeating this procedure, we must continue to find quotient spaces, which makes available new directions to use. Now consider the modified system

$$\dot{g} = (Y + uX + vZ)g$$

Since X and Z can be made arbitrarily large, the drift Y is effectively negligable, making this system compatible with the original system, i.e. the closure of its reachable set is the same as the original system. This new system has two fast directions, X and Z, which we quotient from the convexified adjoint orbit. This is done by computing

$$conv(Ad_{e^{uX+vZ}} Y)/span\{X, Z\}$$

First, it is easy to compute

$$Ad_{e^{uX+vZ}} Y = \{Y + uZ : u \in \mathbb{R}\}$$

Its convex hull quotient the fast X and Z directions is

$$conv(Ad_{e^{uX+vZ}} Y) = \{Y + uZ : u \in \mathbb{R}\}/span\{X, Z\} = \{Y\}$$

This is a singleton, so by Proposition 3.6, the system is not small-time controllable.

3.4 Controllability on $SE(2)$

3.4.1 (Y, X)

$[Y, X] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.4.2 (Z, X)

$[Z, X] = Y$, thus the pair is evidently bracket generating. The convexified adjoint orbit after we quotient out the fast direction X, is

$$conv(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

which is a straight line not passing through the origin. Taking the fast direction into account again, we obtain

$$conv(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily fast by taking large u and v. This effectively makes Z negligable. We check the commutator

$$[Y, X] = 0 \in span\{Y, X\}$$

X and Y were unable to generate a new direction, so we need to continue to find quotient spaces, to obtain new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

that is compatible with the original system (through the same reasoning as before, taking large X and Y to make Z negligable). This system has two fast directions, X and Y, which we quotient from the convexified adjoint orbit. That is,

$$conv(Ad_{e^{uX-vY}}Z)/span\{X, Y\}$$

We first compute

$$Ad_{e^{uX-vY}}Z = \{Z - vX - uY : u, v \in \mathbb{R}\}$$

Convexifying it gives a plane passing through Z that is parallel to the X-Y plane. We then quotient out the fast X and Y directions, giving

$$conv(Ad_{e^{uX-vY}}Z)/span\{X, Y\} = \{Z\}$$

which is a singleton. By Proposition 3.6 the system is not small-time controllable.

3.4.3 (X, Z)

$[X, Z] = -Y$, and so

$$span\{X, Z, [X, Z]\} = \mathfrak{se}(2),$$

meaning the pair is bracket generating. We proceed to compute the convexified adjoint orbit. The adjoint orbit is

$$Ad_{\overline{e^{sZ}}}X = \{\cos(s)X + \sin(s)Y : s \in \mathbb{R}\},$$

and its convex hull quotient the fast direction Z is

$$conv(Ad_{\overline{e^{sZ}}}X)/\mathbb{R}Z = \{(X, Y) : X^2 + Y^2 \leq 1\},$$

which describes the disc bounded by the unit circle in the X-Y plane. Therefore, by Proposition 3.6, the system is not small-time controllable.

3.5 Controllability on $SH(2)$

3.5.1 (X, Z)

$[X, Z] = -Y$, hence the system is bracket generating. One can then check that the convexified adjoint orbit, after we quotient out the fast direction Z, is the region to the right of one branch of a hyperbola in the X-Y plane, given by

$$conv(Ad_{e^{vZ}}X)/\mathbb{R}Z = conv\{\cosh vX + \sinh vY : v \in \mathbb{R}\}$$

Similar to the $(E + F, H)$ case in $SL(2)$, we attempt to apply the barrier function method. Consider the function

$$F = x\cosh\theta - y\sinh\theta$$

Now, let us transform the matrix representation of the system (X, Z) into the state-space representation. From

$$\dot{g} = (X + uZ)g$$

we have

$$\begin{cases} \dot{x} &= uy + 1 \\ \dot{y} &= ux \\ \dot{\theta} &= u \end{cases}$$

Using the above choice of F,

$$F(I) = 0$$

at the identity,

$$\dot{F}(g(t)) = \dot{x}\cosh\theta + x\sinh\theta\dot{\theta} - \dot{y}\sinh\theta - y\cosh\theta\dot{\theta} = \cosh\theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and

$$F(\exp(-\varepsilon X)) = F\left(\begin{pmatrix} 1 & 0 & -\varepsilon \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \ \forall\varepsilon > 0$$

arbitrarily close to the identity. Hence, by Proposition 3.7, (X, Z) is not small-time controllable.

3.5.2 (Y, Z)

$[Y, Z] = -X$, so the pair is bracket generating. One can check that the convexified adjoint orbit, after we quotient out the fast direction Z, is the region above one branch of a hyperbola in the X-Y plane, given by

$$conv(Ad_{e^{vZ}}Y)/\mathbb{R}Z = conv\{\sinh vX + \cosh vY : v \in \mathbb{R}\}$$

Here, the barrier function method is almost identical to the (X, Z) case. First, we write the matrix differential equation

$$\dot{g} = (Y + uZ)g$$

in its state-space representation. That is,

$$\begin{cases} \dot{x} &= uy \\ \dot{y} &= ux + 1 \\ \dot{\theta} &= u \end{cases}$$

Consider the function

$$F(g(t)) = y\cosh\theta - x\sinh\theta$$

for which

$$F(I) = 0$$

at the identity,

$$\dot{F}(g(t)) = \dot{y}\cosh\theta + y\sinh\theta\dot{\theta} - \dot{x}\sinh\theta - x\cosh\theta\dot{\theta} = \cosh\theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and

$$F(\exp(-\varepsilon Y)) = F\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\varepsilon \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \ \forall \varepsilon > 0$$

at a point arbitrarily close to the identity. By Proposition 3.7, the system (Y, Z) is not small-time controllable.

3.5.3 $(X + Y, Z)$

$[X + Y, Z] = -X - Y$, so by Proposition 3.4, the system is not small-time controllable.

3.5.4 $(X - Y, Z)$

$[X - Y, Z] = -Y + X$, so by Proposition 3.4, the system is not small-time controllable.

3.5.5 $(X, X + Y)$

$[X, X + Y] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.5.6 $(X, X - Y)$

$[X, X - Y] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.5.7 $(Z, X + Y)$

$[Z, X+Y] = Y+X = X+Y$, so by Proposition 3.4, the system is not small-time controllable.

3.5.8 $(Z, X - Y)$

$[Z, X - Y] = Y - X$, so by Proposition 3.4, the system is not small-time controllable.

3.5.9 (Z, Y)

$[Z, Y] = X$, therefore this pair is clearly bracket generating. Proceeding, the convexified adjoint orbit after we quotient out the fast direction Y is given by

$$conv(Ad_{e^{vY}} Z)/\mathbb{R}Y = \{Z - vX : v \in \mathbb{R}\},$$

which is a straight line not passing through the origin. Taking the fast direction into account again as in the original system, we have

$$conv(Ad_{e^{vY}} Z) + uY = \{Z + uY - vX : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large controls u and v. We check the commutator

$$[X, Y] = 0 \in span\{X, Y\}$$

X and Y were unable to generate a new direction. We must repeat the above process of finding quotient spaces, attempting to search for new directions to use. Consider the modified system

$$\dot{g} = (Z + uY - vX)g$$

As before, this system is compatible with the original system. This new system has two fast directions, X and Y, which we quotient from the convexified adjoint orbit. That is to say,

$$conv(Ad_{e^{uY-vX}}Z)/span\{X, Y\}$$

We first compute the adjoint orbit to be

$$Ad_{e^{uY-vX}}Z = \{Z - uX + vY : u, v \in \mathbb{R}\}$$

Its convexification is a plane passing through Z parallel to the X-Y plane. We then quotient out the fast X and Y directions, obtaining

$$conv(Ad_{e^{uY-vX}}Z)/span\{X, Y\} = \{Z\},$$

which is a singleton. By Proposition 3.6, the system is not small-time controllable.

3.5.10 (X, Y)

$[X, Y] = 0$, so by Proposition 3.4, the system is not small-time controllable.

3.5.11 (Z, X)

$[Z, X] = Y$, so the system is bracket generating, and we can proceed. The convexified adjoint orbit quotient the fast direction X is a straight line not passing through the origin. It is expressed as

$$conv(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the fast direction into account again as in the original system, we have

$$conv(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Here, the X and Y directions can be made arbitrarily fast by taking the controls u and v to be large. Checking the commutator

$$[Y, X] = 0 \in span\{Y, X\}$$

X and Y were unable to generate a new direction. We need to repeat the process of finding quotient spaces, and search for possible new directions to use. To this end, consider the modified system

$$\dot{g} = (Z + uX - vY)g$$

which is compatible with the original system. This new system has two fast directions, X and Y, which we quotient from the convexified Ad orbit. We would like to determine

$$conv(Ad_{e^{uX-vY}}Z)/span\{X, Y\}$$

We compute the adjoint orbit first.

$$Ad_{e^{uX-vY}}Z = \{Z + vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z that is parallel to the X-Y plane. We then quotient out the fast X and Y directions, yielding

$$conv(Ad_{e^{uX-vY}}Z)/span\{X, Y\} = \{Z\},$$

a singleton. By Proposition 3.6, the system is not small-time controllable.

3.5.12 (Y, X)

$[Y, X] = 0$, so by Proposition 3.4, the system is not small-time controllable.

4 Conclusion

Throughout this paper, we first classified all control systems of the form 3 on all 3D unimodular matrix Lie groups, and then determined their small-time controllability. 22 equivalent systems were found in total, of which only one, the system (F, E) on $SL(2)$, was small-time controllable.

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